

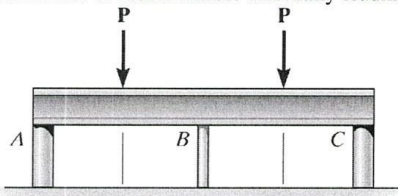
Solid Mechanics (EMCH260 Section 002, Fall 2021)

Mid-Term Exam 2, 8:30 AM, Monday, Oct. 18th 2021

Name (First/Last): Jacob Vaughn Score: _____/100

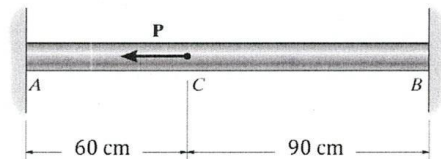
1. Single choice questions. There is one and only one correct answer for each question. ($10 \times 3' = 30'$ in total)

Q1: The rigid beam is supported by three posts. If we need to know the stress in each post, is this a determinate or indeterminate uniaxially loading problem?



- A. Determinate
B. Indeterminate

Q2: The bar is fixed at two ends and the axial load P is applied to C . What compatibility condition can you propose to solve this problem?



- A. $\delta_{AC} = \delta_{CB}$
B. $\delta_{AC} + \delta_{CB} = 0$

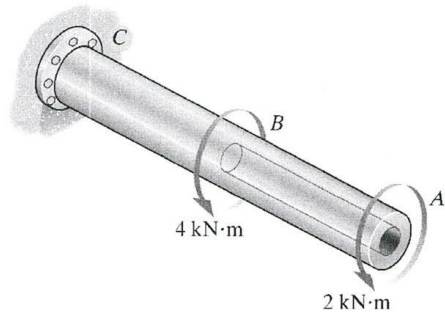
Q3: What type of stress is developed in the cross-section perpendicular to the longitudinal axis of a shaft when external torque is applied?

- A. Normal stress
B. Shear stress

Q4: If a solid shaft has the same diameter as the outer diameter of a hollow or tubular shaft, which one has higher polar moment of inertia?

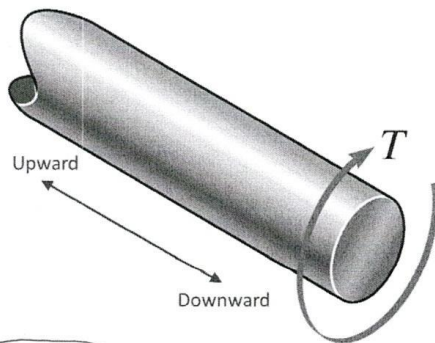
- A. The solid shaft
B. The hollow or tubular shaft

Q5: The shaft is hollow from A to B and solid from B to C . The entire shaft is fixed to the wall at end C . If we need to determine the reaction at end C , is this a determinate or indeterminate torsional problem?



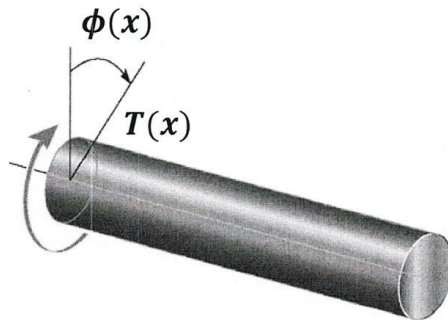
- A. Determinate
- B. Indeterminate

Q6: Using the right-hand-rule, determine the direction the external torque T is pointing to:



- A. Upward
- B. Downward

Q7: Using the right-hand-rule and sign convention, the sign of the internal resultant torque $T(x)$ is:



A. Negative

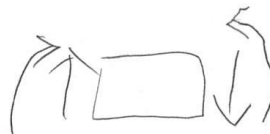
B. Positive

Q8: What type of stress is involved in the flexure formula for a bending problem?

A. Normal stress

B. Shear stress

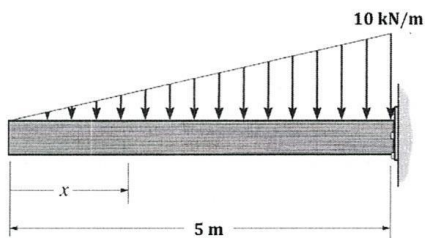
Q9: Use beam sign convention to determine the sign of the shear force shown below (the blue region is a segment of the beam):



A. Negative

B. Positive

Q10: Determine the total equivalent force of the external distributed forces:



A. 25 kN

B. 50 kN

PROBLEM #2

BUILDING MAIN EQUATIONS:

$$\text{Inertia or } J = \frac{\pi}{2}(r)^4 \text{ or if hollow: } J = \frac{\pi}{2}(R_{\text{outer}}^4 - R_{\text{inner}}^4)$$

$$\text{Shear stress or } \tau = \frac{T * r}{J}; \text{ Angle } = \phi = \frac{TL}{JG}$$

$$\phi_{mg} = \phi_{st}$$

$$T = T_{st} + T_{mg}$$

$$J_{st} = \frac{\pi}{2} * 0.02^4$$

$$J_{mg} = \frac{\pi}{2} * (0.04^4 - 0.02^4)$$

SOLVING SAID EQUATIONS

$$\frac{T_{mg}}{\frac{\pi}{2} * 0.02^4 * 20 * 10^9} = \frac{T_{st}}{\frac{\pi}{2} * (0.04^4 - 0.02^4) * 80 * 10^9} \rightarrow T_{st} = 60 T_{mg}$$

$$60T_{mg} + T_{mg} = T \rightarrow T_{mg} = \frac{T}{61}$$

$$\frac{1}{60} * T_{st} + T_{st} = T \rightarrow T_{st} = \frac{60}{61} T$$

MAX TORQUE FOR EACH MATERIAL

$$T = \frac{\tau * J}{r}$$

$$T_{st \text{ MAX}} = \frac{75 * 10^6 * \frac{\pi}{2} * (0.04^4 - 0.02^4)}{0.04} = 2250 * \pi$$

$$T_{mg \text{ MAX}} = \frac{45 * 10^6 * \frac{\pi}{2} * 0.02^4}{0.02} = 180 * \pi$$

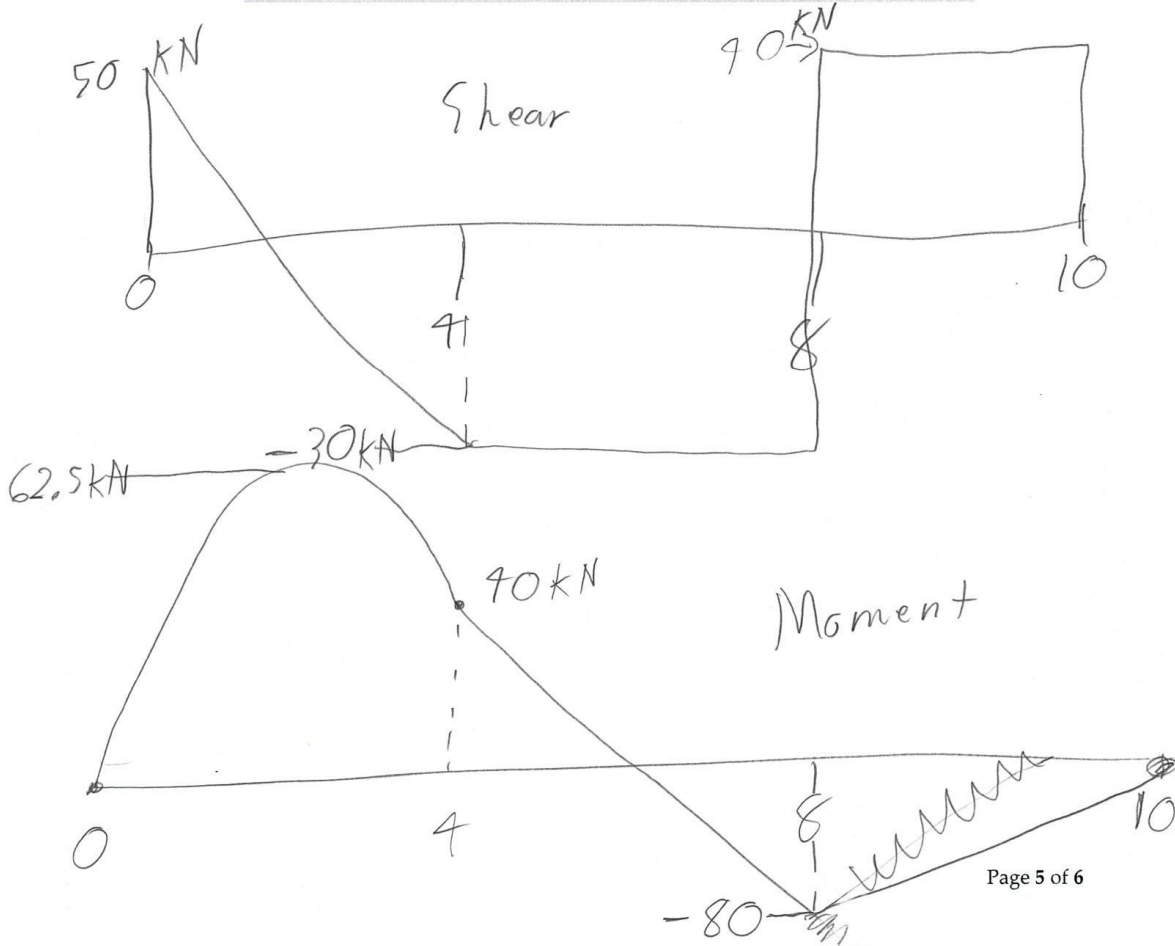
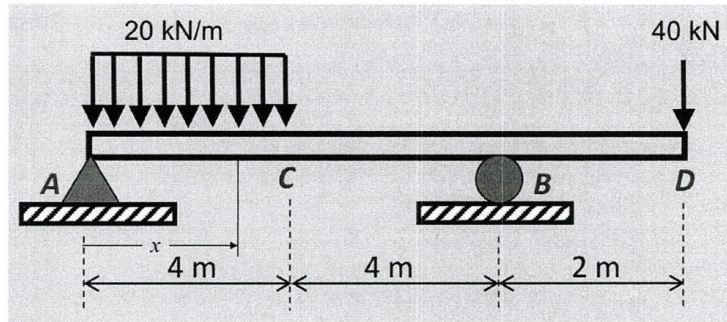
$$T \text{ based on Steel } T_{max} \rightarrow 2250 * \pi = \frac{60}{61} T \rightarrow T = \frac{4575}{2} \pi \cong 7186.4 N \cdot m$$

$$T \text{ based on Magnesium } T_{max} \rightarrow 180\pi = \frac{1}{61} T \rightarrow 10980\pi \cong 34494.7 N \cdot m$$

Steel $T_{max} <$ Magnesium, thus **7186.4 N · m** is the Maximum torque.

$$Angle = \phi = \frac{T_{st}L}{J_{st}G_{st}} = \frac{2250 * \pi * 0.9}{\frac{\pi}{2} * 0.02^4 * 80 * 10^9} = \frac{81}{265} \cong 0.316 \text{ radians or } 18.13 \text{ degrees.}$$

3. Determine the shear and moment diagrams for the beam and plot the diagrams. Also determine the absolute maximum normal stress in the beam with external loading shown below. The beam has a uniform square cross-section with lateral size $a = 0.2$ m. Note: the shear and moment diagrams can be calculated either by section method or graphical method. (40' in total)



$$I = \frac{1}{12} b h^3$$

$$I = \frac{1}{12} (0.2)(0.2)^3 \rightarrow 0 = \frac{M_c}{I} \rightarrow \frac{-80 \cdot 0.1 \text{ m}}{\frac{1}{12} (0.2)(0.2)^3}$$

$$\boxed{-60 \text{ MN/m}^2}$$